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S.E. (Civil) (II Sem.) EXAMINATION, 2019

FLUID MECHANICS-I

(2015 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Answer Q. 1 or Q. 2, Q. 3 or Q. 4, Q. 5 or Q. 6, Q. 7 or Q. 8.

(ii) Answers to the all questions should be written in single answer book.

(iii) Neat diagram must be drawn wherever necessary.

(iv) Figures to the right indicate full marks.

(v) Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator (non-programmable) and steam tables is allowed.

(vi) Assume suitable data, if necessary.

1. (a) Explain in brief :

(i) Rheological Diagram, and

(ii) Newtonian and non-Newtonian Fluid.

[3+3]

P.T.O.

- (b) Figure 1 (b) shows a manometer connected to a pipeline, containing an oil of specific gravity 0.8. Find the pressure of oil in the pipe. [6]

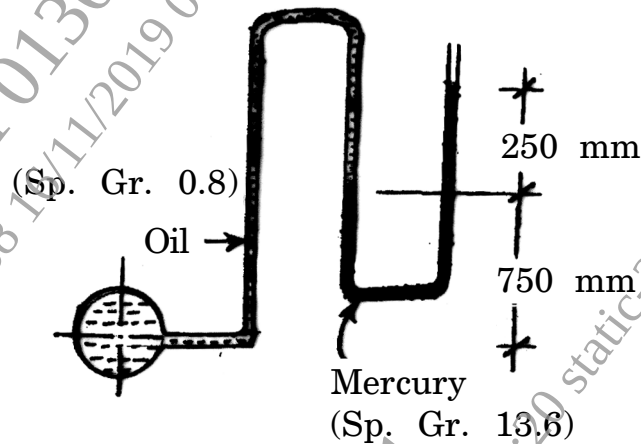


Fig. 1 (b)

Or

2. (a) What do you mean by dimensional homogeneity ? Explain the procedure for solving problems by Buckingham's π Theorem. [6]
- (b) A wooden block of specific gravity 0.75 floats in water with its 0.4 m side vertical. If the size of the block is 1 m $\times$ 0.5 m $\times$ 0.4 m, find its metacentric height. [6]

3. (a) Derive the Bernoulli's Equation with usual notations. [6]

- (b) The velocity components in a fluid flow are given by :

$$u = 2xy; v = a^2 + x^2 - y^2$$

- (i) Show that the flow is possible
- (ii) Derive the relative stream function. [6]

Or

4. (a) Derive the continuity equation for one-dimensional flow along a stream line with usual notations. [6]

- (b) A horizontal venturimeter with inlet diameter 200 mm and throat diameter 100 mm is used to measure the flow of water. The pressure at inlet is 0.18 N/mm^2 and the vacuum pressure at the throat is 280 mm of mercury. Find the rate of flow. The value of C_d may be taken as 0.98. [6]
5. (a) A fluid of viscosity 8 poise and specific gravity 1.2 is flowing through a circular pipe of diameter 100 mm. The maximum shear stress at the pipe wall is 210 N/m^2 . Find :
- The pressure gradient,
 - The average velocity, and
 - Reynolds number of the flow. [7]
- (b) Explain with neat sketches "Boundary Layer Separation and its control". [6]

Or

6. (a) In case of flow of viscous fluid through circular pipe; show that the loss of pressure head is given by the following expression : [6]

$$\frac{p_1 - p_2}{\rho g} = \left[\frac{32 \mu \bar{u} L}{\rho g D^2} \right].$$

- (b) Find the momentum thickness, displacement thickness and energy thickness for velocity distribution in the boundary layer given by $\frac{u}{U} = \frac{y}{\delta}$, where u is the velocity at a distance y from the plate and $u = U$ at $y = \delta$, where δ boundary layer thickness. Also calculate the value of $\frac{\delta^*}{\theta}$. [7]

7. (a) A pipe of 100 mm diameter is carrying water. If the velocities at the pipe centre and 30 mm from the pipe centre are 2.0 m/s and 1.5 m/s respectively and flow in the pipe is turbulent, calculate the shear friction velocity and wall shearing stress. [6]
- (b) The main pipe divides into two parallel pipes which again forms one pipe. The data is as follows :
 First parallel pipe : Length = 1000 m, Diameter = 0.8 m
 Second parallel pipe : Length = 1000 m, Diameter = 0.6 m
 Coefficient of friction for each parallel pipe = 0.005
 If the total rate of flow in the main is 2 m³/s, find the rate of flow in each parallel pipe [7]

Or

8. (a) Derive the following expression with usual notations for loss of head in pipes due to sudden enlargement. [7]

$$h_e = \frac{(V_1 - V_2)^2}{2g}.$$

- (b) Explain the “Prandtl’s mixing length theory” for turbulent shear stress. [6]